Savitur Swarup

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Education

University of Pennsylvania — Philadelphia, PA

B.S.E. Computer Science (SEAS); B.S. Economics, Finance (Wharton); GPA: 3.8/4.0 — *Expected May 2028*

- **Coursework:** Data Structures & Algorithms, Computer Systems, Automata & Computability, Discrete Math, Probability, Linear Algebra, Multivariable Calculus
- **Awards:** USA Computing Olympiad Gold Division; American Math Olympiad Gold Medalist

Experience

Software Engineer — July 2026 – Present

Tara Ventures — *Singapore*

- Built the company's first ad-optimization pipeline (Python, Meta Marketing API, Google Ads API): A/B-tested LLM creative across verticals with a bandit-clustering feedback loop, accelerating rollout and lifting campaign performance.
- Implemented multi-armed bandit budget allocation with automatic underperformer pruning; took ad-driven revenue from zero to a **5:1 ROAS across 30 campaigns**.

Co-Founder & Software Engineer — Nov. 2025 – Present

Bridge AI (Remote) — *Philadelphia, PA*

- Built an agentic system executing real CRM writes across a 45-tool registry: **87.2% tool-selection accuracy** (41/47 labeled tasks), with all **22 state-mutating capabilities** confirm-gated or undo-backed, enforced as a load-time invariant.
- Engineered a routing layer across 7 LLM providers with per-task routing, fallback chains, and per-call cost attribution; a dual-judge benchmark exposed misrouted traffic, and rerouting **cut per-task inference cost 73%** at tied quality.
- Built a hybrid RAG pipeline on PostgreSQL/pgvector (halfvec embeddings, tsvector search, async embedding worker); fp16 quantization **halved vector storage** (12,288→6,144 B/vector) at no measured recall cost.

Research Intern — May 2026 – Jul. 2026

RWTH Aachen University, Forschungszentrum Jülich — *Jülich, Germany*

- Harmonized **8 public EEG datasets** (504 subjects) into one pipeline (Python, MNE); microstate features flagged Alzheimer's at **AUC 0.89**, and the Alzheimer's-trained model transferred to unseen FTD at **AUC 0.92**.
- Ran robustness checks (age-confound regression, 5000-shuffle permutation tests, cross-site validation, ComBat harmonization), **catching a spectral confound** that had inflated a false depression result.

Founder — Jan. 2023 – Aug. 2025

Jiro Web Solutions — *Singapore*

- Founded a web development agency for small businesses across Asia; hired and trained a team of 5, shipping **20+ sites and mobile apps** (React, React Native, Vue) with auth, Stripe payments, and Supabase/Firebase backends.

Research Intern — Jun. 2024 – Sep. 2024

*A*STAR, Centre for Frontier AI Research* — *Singapore*

- Benchmarked **10 LLMs across 20 datasets** on multimodal sarcasm detection (accuracy, F1, precision, recall) under Prof. Cheston Tan; fine-tuned a custom model on the findings to improve context understanding and multimodal integration.

Projects

nanoserve — 2026

LLM Inference Server | Python, PyTorch, Triton, C++, CUDA — *github.com/Savi-Swar/nanoserve*

- Built an LLM inference server from scratch (continuous-batching scheduler, paged KV allocator, custom Triton attention kernel, CUDA-graph capture): **24→578 tok/s** (**23.7**×) on one GPU, **86% of vLLM's throughput** at lower p99 latency.
- Sharded a **7B model exceeding single-GPU memory** across two GPUs at **7.8**× naive throughput; ported the scheduler to **C++**, proven equivalent by **hashed replay** of every scheduling decision.
- Chaos-tested cancellation under load with **zero leaked KV blocks**; reported the same bug class in **vLLM's V1 scheduler** (#49684); a cost model committed before measurement **predicted a 6× larger model's performance within 5.4%**.

Vig — 2026

Automated Research & Data Platform | Python, pandera, GitHub Actions — *github.com/Savi-Swar/desk*

- Distributed a 20-hour rate-limited backfill across a **12-worker CI matrix** (partitioning invariants pinned by tests, idempotent merge): **74k records in 70 min**, 12/12 first run; cross-feed reconciliation caught a **1,735**× overcount.
- Wrote a **C++17** parallel stream-reconstruction engine: **2.0M msg/s**, **8**× the Python reference, output **parity-pinned 246/246**; phase profiling isolated a memory-bandwidth wall and a **2.8**× serialization fix.

Technical Skills

Languages: Python, TypeScript, JavaScript, C/C++, Java, SQL
ML & Data: PyTorch, Hugging Face, pandas, NumPy, scikit-learn, pgvector, MNE-Python
Backend & Web: PostgreSQL, Node.js, Express, Next.js, React, React Native, asyncio, REST/SSE
Infra & Testing: Docker, GitHub Actions CI, pytest, pandera, Prometheus, Supabase, Firebase